National University of Computer and Emerging Sciences
Lahore Campus

Differential Equation (MT1006)

Sessional-II Exam

Date: April 4th 2024

Course Instructor(s)

Dr. Tayyaba Naz

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Roll No ________ Section ________ Student Signature ________

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Attempt all the questions.

CLO #2: Determine the solutions of 1st order DEs and higher order DEs

Q1: (a) Determine the solution of the given differential equation by using undetermined coefficients approach. **[10 marks]**

$x_1 = 3$

$x_2 = -1$

$$y'' - 2y' - 3y = 65\,x\cos 2x$$

Q1: (b) Solve the given differential equation using method reduction of order, while the indicated function $y_1(x)$ is a solution of the associated homogeneous equation. **[10 marks]**

$-\frac{1}{5}x^{-3}$

$$x^2y'' + 2xy' - 6y = 0\ ; \qquad y_1(x) = x^2$$

Q1: (c) Verify that the given functions form a fundamental set of solutions of the differential equation on the indicated interval. Also form the general solution. **[10 marks]**

$$y'' - 2y' + 5y = 0\ ; \qquad y_1(x) = e^x\cos 2x, \quad y_2(x) = e^x\sin 2x \;\; on\ (-\infty, \infty).$$

CLO #3: Formulate DEs with applications

Q2: The sugar-making process contains a step called "inversion" in which cane sugar is dissolved in water. The rate at which the quantity of unconverted sugar is changing is proportional to the amount present. If 550 kg are present initially and 400 kg are present after 10 hours, how much is left after 15 hours? At what rate is the unconverted sugar changing after 15 hours? **[10 marks]**

341.7 10

$4xe^{2x}$

$(Ax+B)e^{2x}$

$A\cos 2x + A\sin 2x.$

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